

sintaxismark — Documentation

Syntactic annotation over running text in L^AT_EX

v0.6.21 · 2026-06-15

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1 Introduction

`syntaxismark` is a $\text{\LaTeX}$ package for marking syntactic constituents and grammatical functions directly over running text, such as grammar examples, corpus sentences, and classroom materials, without having to draw syntactic trees. It generates visual annotations with `TikZ`: lower open brackets (*open boxes*), upper span lines, labels above or below the text (`\tagabove/\tagbelow`), and underlining (`\synul`).

The package is designed to work well with `gb4e`, but it can also be used in ordinary running text. Since it relies on `TikZ`'s `remember picture` mechanism, the document must normally be compiled twice for the annotations to be placed correctly.

1.1 Dependencies

- `tikz`, with the `calc` and `arrows.meta` libraries
- `xparse`
- `xcolor`
- `expl3`
- `varwidth`, for line-wrapping support

2 Loading the package and visual schemas

```
\usepackage[<schema>]{syntaxismark}
```

The optional schema sets global visual parameters for different use contexts. The available schemas are:

<code>docencia</code>	Thicker lines and wider spacing. Useful for teaching materials, handouts, and classroom-oriented examples.
<code>impresion</code>	Thin lines and compact spacing. Suitable for printed material and dense handouts.
<code>diapositivas</code>	Thick lines, larger spacing, and larger arrowheads. Optimized for Beamer slides or similar presentations.
<code>publicacion</code>	Very thin lines and compact layout. Suitable for academic articles and publication-style examples.

A schema can also be applied anywhere in the document with:

```
\setsyntaxisschema{diapositivas}
```

If no package option is specified, the default values of the keys listed in Section 3 are used.

3 Configuration keys

All drawing commands accept an optional argument with `pgfkeys` from the `/syntaxis` key family. Keys may also be set globally with `\pgfkeys{/syntaxis,...}`.

Key	Default	Description
color	<i>black</i>	Color of lines and labels. Accepts any <code>xcolor</code> color.
liney	<i>1ex</i>	Height of the upper horizontal line in <code>\spanline</code> .
linegap	<i>5.5pt</i>	Space between the text and the upper line.
labsep	<i>1ex</i>	Space between the line and its label in span lines.
linewidth	<i>0.6pt</i>	Line width.
boxd	<i>3ex</i>	Base depth of the lower line in <code>\openbox</code> .
boxsep	<i>1.0ex</i>	Space between the lower line and its label in open boxes.
labelsep	–	Per-instance label–line separation. Overrides <code>boxsep</code> .
labelsepboxes	–	Global label–line separation for nested open boxes.
nestedgap	<i>0.3ex</i>	Additional vertical space between nested open boxes or nested lower labels.
labheight	<i>1.1ex</i>	Vertical space reserved for the label.
tagy	<i>1.8ex</i>	Vertical offset for <code>\tagabove</code> / <code>\tagbelow</code> .
tagx	<i>0pt</i>	Horizontal offset for <code>\tagbelow</code> .
underw	<i>0.8pt</i>	Width of the underline drawn by <code>\synul</code> .
labelalign	<i>center</i>	Label alignment: <i>center</i> , <i>left</i> , or <i>right</i> .
autogap	<i>0.6ex</i>	Extra automatic spacing used by <code>autodepth</code> .
leftend	<i>true</i>	Draw the left vertical end of the bracket.
rightend	<i>true</i>	Draw the right vertical end of the bracket.
autodepth	<i>true</i>	Automatically compute the depth needed by nested open boxes.
leftarrow	<i>false</i>	Add an arrowhead at the left end of the horizontal line.
rightarrow	<i>false</i>	Add an arrowhead at the right end of the horizontal line.
arrowlen	<i>1.3ex</i>	Arrowhead length.
arrowtip	<i>Latex</i>	Arrowhead type, e.g. <code>Latex</code> , <code>Stealth</code> , etc. See <code>arrows.meta</code> .
arrowscale	<i>1</i>	Arrowhead scale.
arrowinset	<i>0pt</i>	Arrowhead inset.

4 Primitive commands

The following low-level commands are the building blocks on which the predefined shortcuts are based.

4.1 `\openbox` — lower open bracket with label

```
\openbox[<keys>]{<label>}{<content>}
```

Draws an open bracket below the content, with the label centered below the lower horizontal line. The label is typeset automatically in small caps.

The bracket consists of a left vertical end, a right vertical end, and a lower horizontal line. The vertical ends are controlled by the `leftend` and `rightend` keys.

Example:

```
\begin{exe}  
  \ex \openbox{od}{a Juan}  
\end{exe}
```

(1) 

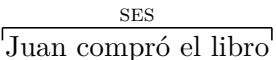
4.2 `\spanline` — upper span line with label

```
\spanline[<keys>]{<label>}{<content>}
```

Draws a bracket above the content, with the label above the horizontal line. This is useful for subjects, predicates, clauses, or any constituent whose span should be marked above the text.

Example:

```
\begin{exe}  
  \ex \spanline{ses}{Juan compró el libro}  
\end{exe}
```

(2) 

4.3 `\tagabove` and `\tagbelow` — floating labels

```
\tagabove[<keys>]{<label>}{<content>}  
\tagbelow[<keys>]{<label>}{<content>}
```

Place a small-caps label above or below the content without drawing a line. The vertical position is controlled by `tagy`; the horizontal offset is controlled by `tagx`.

Example:

```
\begin{exe}
\ex \tagbelow{n}{Juan} vio \tagbelow{n}{a María}
\end{exe}
```

(3) Juan vio a María
N N

4.4 \synul — underline

```
\synul[<keys>]{<content>}
```

Draws an underline below the content. Its width is controlled by `underw`. It is often combined with `\tagbelow`; see `\NN`, `\NV`, and related shortcuts.

Example:

```
\begin{exe}
\ex \synul{Juan}
\end{exe}
```

(4) Juan

5 Predefined shortcuts

The package defines shortcuts for common syntactic functions and grammatical categories. The basic interface is:

```
\COMMAND[<keys>]{<content>}
```

5.1 Dual convention: \tagbelow vs. \openbox

Since v0.6.19, most shortcuts for syntactic functions and modifiers have two forms, depending on whether the starred form is used:

- **Unstarred form:** equivalent to `\tagbelow`; it only places a label below the content, without drawing a bracket.
- **Starred form (`\COMMAND*`):** equivalent to `\openbox`; it draws the lower bracket and its label.

```
\COMMAND[<keys>]{<content>} % = \tagbelow{<label>}{<content>}
\COMMAND*[<keys>]{<content>} % = \openbox{<label>}{<content>}
```

Example:

```
\begin{exe}
\ex \OD{el libro} \quad \OD*{el libro}
\end{exe}
```

(5) $\frac{\text{el libro}}{\text{OD}}$ $\boxed{\text{el libro}}$
OD OD

This convention applies to `\OD`, `\OI`, `\CCL`, `\COMP`, `\MD`, `\MI`, `\CAg`, `\CREG`, `\CSim`, `\CL`, `\CM`, `\PSO`, `\PSNO`, `\POO`, `\PONO`, `\ACT`, `\ACL`, `\ACM`, `\ACF`, `\ACC`, `\ACCant`, `\ACI`, `\ACFin`, `\MG`, `\Ap`, `\MF`, `\IPCR`, and `\Incl`. The commands `\N`, `\NP`, `\NA`, `\NAD`, the underlined variants (`\NN`, `\NV`, `\MGV`, etc.), and the `\spanline` shortcuts (`\SES`, `\ST`, etc.) do **not** have a starred form.

5.2 Syntactic functions: dual `\tagbelow`/`\openbox` form

By default, these commands behave like `\tagbelow`. With `*`, they draw the lower `\openbox` bracket as well.

Command	Label	Function
<code>\OD</code>	OD	Direct object
<code>\OI</code>	OI	Indirect object
<code>\CCL</code>	CCL	Locative circumstantial complement
<code>\COMP</code>	COMP	Complement
<code>\MI</code>	MI	Internal modifier
<code>\CAg</code>	CAG	Agent complement
<code>\CREG</code>	CREG	Prepositional/governed complement
<code>\CSim</code>	CSIM	Similar complement
<code>\CL</code>	CL	Clause or clause-like unit
<code>\CM</code>	CM	Manner/modal complement
<code>\PSO</code>	PSO	Object-oriented secondary predicate
<code>\PSNO</code>	PSNO	Non-oriented secondary predicate
<code>\POO</code>	POO	Object-oriented predicate
<code>\PONO</code>	PONO	Non-obligatory oriented predicate
<code>\ACT</code>	ACT	Temporal circumstantial adjunct
<code>\ACL</code>	ACL	Locative circumstantial adjunct
<code>\ACM</code>	ACM	Manner circumstantial adjunct
<code>\ACF</code>	ACF	Frequency circumstantial adjunct
<code>\ACC</code>	ACC	Cause circumstantial adjunct
<code>\ACCant</code>	ACCANT	Fronted circumstantial adjunct
<code>\ACI</code>	ACINST	Instrumental circumstantial adjunct
<code>\ACFin</code>	ACFIN	Purpose circumstantial adjunct

5.3 Syntactic functions with `\spanline`: upper brackets

Command	Label	Function
<code>\SES</code>	SES	Simple overt subject
<code>\ST</code>	ST	Null/tacit subject
<code>\SEC</code>	SEC	Compound overt subject
<code>\PVS</code>	PVS	Simple verbal predicate
<code>\PVC</code>	PVC	Compound verbal predicate
<code>\OS</code>	OS	Simple clause/sentence

5.4 Lower labels without a line

The commands `\MD`, `\MG`, `\Ap`, `\MF`, `\IPCR`, and `\Incl` also accept the starred `\openbox` form. The commands `\N`, `\NP`, `\NA`, and `\NAD` exist only as `\tagbelow` shortcuts.

Command	Label	Note
<code>\MD</code>	MD	Head modifier
<code>\N</code>	N	Head, without underline
<code>\NP</code>	NP	Head of a prepositional phrase
<code>\NA</code>	NA	Head of an adjectival phrase
<code>\NAD</code>	NADV	Head of an adverbial phrase
<code>\MG</code>	MG	Degree modifier
<code>\Ap</code>	AP	Apposition
<code>\MF</code>	MF	Phrasal modifier, with <code>tagx=0.5ex</code>
<code>\IPCR</code>	IPCR	Inanimate/post-copular referent, or local label
<code>\Incl</code>	INCL	Parenthetical/incise

5.5 Labels combined with underlining

The following commands combine `\tagbelow` and `\synul`, marking the head with both an underline and a label.

Command	Equivalent to
<code>\NN</code>	<code>\tagbelow{nn}{\synul{...}}</code>
<code>\NV</code>	<code>\tagbelow{nv}{\synul{...}}</code>
<code>\MGV</code>	<code>\tagbelow{mg}{\synul{...}}</code>
<code>\ApV</code>	<code>\tagbelow{ap}{\synul{...}}</code>
<code>\MFV</code>	<code>\tagbelow{mf}{\synul{...}}</code>
<code>\IPCRV</code>	<code>\tagbelow{ipcr}{\synul{...}}</code>
<code>\InclV</code>	<code>\tagbelow{incl}{\synul{...}}</code>

6 The oracion environment

```
\begin{oracion}[<options>]
...
\end{oracion}
```

This environment presents an analyzed sentence outside `gb4e`. It adds vertical space above and below the sentence.

Example:

```
\begin{oracion}
  \SES{Juan} \PVS{\NV{compró} \OD{el libro}}
\end{oracion}
```

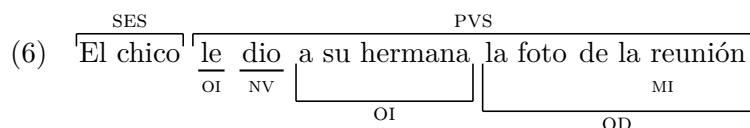
$$\overbrace{\text{Juan}}^{\text{SES}} \overbrace{\text{compró el libro}}^{\text{PVS}}$$

$$\text{NV} \quad \text{OD}$$

7 Nesting

Commands may be nested. With `autodepth=true`, the default, the package automatically computes the depth needed by parent open boxes when they contain nested child boxes or lower labels, avoiding visual overlap.

```
\begin{exe}
\ex \SES{El chico} \PVS{\OI{le} \NV{dio} \OI*{a su hermana}
\OD*{la foto \MI{de la reunión}}}
\end{exe}
```



To disable automatic depth computation and set the depth manually:

```
\openbox[autodepth=false, boxd=5ex]{od}{...}
```

8 Long sentences and line wrapping

When a sentence exceeds the line width, the package provides three complementary mechanisms.

8.1 `\SMlinewrap`

```
\SMlinewrap{<content>}
```

Wraps the content in a `\linewidth`-wide container that allows natural line breaks. It is intended for use inside `\ex` examples from `gb4e`.

```
\begin{exe}
\ex \SMlinewrap{El \openbox{mi}{departamento de los monoblocks
deteriorados de tres pisos de color bordó} quedaba en la planta baja.}
\end{exe}
```

8.2 `\openboxNB` and `\spanlineNB`

```
\openboxNB[<keys>]{<label>}{<content>}
\spanlineNB[<keys>]{<label>}{<content>}
```

These are no-break variants. They wrap the content in `\mbox`, preventing the annotated span from being split across two lines. This keeps the TikZ anchors on the same line and prevents the bracket line from becoming slanted.

8.3 `\breakspan`

```
\breakspan{<line-1>}{<line-2>}
```

For cases in which a span must cross an explicit line break, `\breakspan` automatically applies:

- to the first argument: `rightarrow=true`, `rightend=false`;
- to the second argument: `leftarrow=true`, `leftend=false`.

The effect is propagated to all levels of nesting inside each argument.


```
\breakspan
{\SES{\MI{El departamento de unos monoblocks}}}
{\SES{\MI{quedaba en la planta baja.}}}
```

9 Arrowheads and bracket ends

The `\leftarrow` and `\rightarrow` keys add arrowheads to the horizontal line in both `\openbox` and `\spanline`. The `\leftend` and `\rightend` keys control whether the vertical ends of the bracket are drawn.

```
\begin{exe}
\ex \openbox[rightarrow,rightend=false]{od}{el libro}
\end{exe}
```

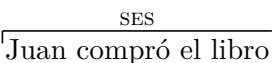
(7) 

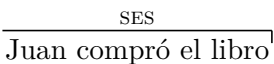
The end keys may be used independently to suppress only one vertical end:

```
\begin{exe}
\ex \openbox[rightend=false]{od}{el libro}
\ex \openbox[leftend=false]{od}{el libro}
\ex \spanline[rightend=false]{ses}{Juan compró el libro}
\ex \spanline[leftend=false]{ses}{Juan compró el libro}
\end{exe}
```

(8) 

(9) 

(10) 

(11) 

Combining both arrow keys gives arrows on both ends, with or without vertical ends:

```
\begin{exe}
\ex \openbox[leftarrow,rightarrow,leftend=false,rightend=false]{od}{el libro}
\ex \spanline[leftarrow,rightarrow,leftend=false,rightend=false]{ses}{Juan compró
el libro}
\end{exe}
```

(12) 

(13) $\overleftrightarrow{\text{SES}}$ Juan compró el libro

If both arrowheads and vertical ends are omitted, the result is a plain horizontal line:

```
\begin{exe}
  \ex \openbox[leftend=false,rightend=false]{od}{el libro}
  \ex \spanline[leftend=false,rightend=false]{ses}{Juan compró el libro}
\end{exe}
```

(14) el libro
 $\overline{\text{OD}}$

(15) $\overline{\text{SES}}$ Juan compró el libro

The arrowhead type is configured with `arrowtip`. Any `arrows.meta` arrowhead can be used, such as `Latex`, `Stealth`, or `Triangle`. Size is controlled with `arrowlen` and `arrowscale`.

10 Colors

```
\openbox[color=red!70!black]{od}{el libro}
\spanline[color=blue]{ses}{Juan comió}
\tagbelow[color=teal]{n}{Juan}
```

(16) el libro
 $\overline{\text{OD}}$

$\overline{\text{SES}}$
Juan comió
Juan
 N

11 Compact mode

The `.sty` file includes, enabled by default, an “extreme compact mode” block at the end of the file. It replaces dynamic vertical reservation with a fixed, minimal reservation:

```
\renewcommand*\syn@reservebelow[1]{\vadjust{\kern 4.5ex}}
\renewcommand*\syn@reserveabove[1]{%
  \leavevmode\vrule height 7.5ex depth 0pt width 0pt\relax}
```

This reduces vertical whitespace in dense documents, at the cost of removing full automatic adjustment for very deep nested spans. If deeply nested open boxes require more space, adjust the `4.5ex/7.5ex` values or remove that block from the package file.

12 Typical use with gb4e

```
\usepackage{gb4e}\noautomath
\usepackage{sintaxismark}
```

```

\begin{exe}
  \ex \SES{María} \PVS{\NV{compró} \OD{un libro \MI{de poesía}}}

  \ex \SES{Los chicos}
    \PVS{\OI{le} \NV{dieron} \OI*{a la profesora}
      \OD*{las notas}}

  \ex[*]{ \openbox{od}{\tagbelow{n}{Ese} libro} cayó }

  \ex \SMlinewrap{%
    El \MI{departamento de los monoblocks deteriorados}
    \NV{quedaba} \ACL{en la planta baja}.}
\end{exe}

```

(17) $\overbrace{\text{María}}^{\text{SES}} \overbrace{\underbrace{\text{compró}}_{\text{NV}} \underbrace{\text{un libro de poesía}}_{\text{OD}}}_{\text{MI}}^{\text{PVS}}$

(18) $\overbrace{\text{Los chicos}}^{\text{SES}} \overbrace{\underbrace{\underbrace{\text{le}}_{\text{OI}} \underbrace{\text{dieron}}_{\text{NV}}}_{\text{OI}} \underbrace{\text{a la profesora}}_{\text{OD}}}_{\text{PVS}} \underbrace{\text{las notas}}_{\text{OD}}$

13 Command summary

Command	Type	Short description
\openbox	primitive	Lower open bracket + label
\spanline	primitive	Upper span line + label
\tagabove	primitive	Floating label above
\tagbelow	primitive	Floating label below
\synul	primitive	Underline
\OD, \OI, \CCL, \COMP	shortcut*	Dual direct/indirect object and complement labels
\MI	shortcut*	Dual internal modifier
\CAg, \CREG, \CSim	shortcut*	Dual agent/governed/similar complement labels
\CL, \CM	shortcut*	Dual clause/modal complement labels
\PSO, \PSNO, \POO, \PONO	shortcut*	Dual secondary-predicate labels
\ACT, \ACL, \ACM, \ACF	shortcut*	Dual circumstantial-adjunct labels
\ACC, \ACCant, \ACI, \ACFin	shortcut*	More dual circumstantial-adjunct labels
\SES, \ST, \SEC	shortcut	Span-line subject labels
\PVS, \PVC, \OS	shortcut	Span-line predicate/sentence labels
\MD	shortcut*	Dual head modifier
\N, \NP, \NA, \NAD	shortcut	Lower head labels
\MG, \Ap, \MF, \IPCR, \Incl	shortcut*	Dual modifier/apposition/parenthetical labels
\NN, \NV	shortcut	Lower label + underline for nominal/verbal heads
\MGV, \ApV, \MFV, \IPCRV, \InclV	shortcut	Lower label + underline variants
\openboxNB	wrapper	No-break open box
\spanlineNB	wrapper	No-break span line
\SMlinewrap	wrapper	Line-wrapping container

Command	Type	Short description
<code>\breakspan</code>	wrapper	Explicitly split span across two lines
<code>\setsintaxischema</code>	configuration	Apply a visual schema

`shortcut*` means that the command has a starred form equivalent to `\openbox`; the unstarred form is equivalent to `\tagbelow`.

14 Technical notes

- The package uses TikZ’s `remember picture` mechanism: **two compilation passes** are normally required for correct positioning.
- All labels are automatically typeset in small caps with `\textsc`.
- Each annotation instance increments an internal counter (`syn@id`) to generate unique TikZ node names. This makes repeated uses of the same command robust within the same document.
- Nested `\openbox` and `\tagbelow` instances propagate their maximum depth to the parent through the `\syn@reportdepth` mechanism, which is inherited through TeX groups and accumulated in `\syn@childmaxd`. As a result, a nested lower label, for example `\MI` inside `\OD`, is drawn lower than its container’s label, avoiding overlap.
- `\breakspan` works because `syn@breakmode` is a TeX counter whose value is inherited by child groups. Once the value is set before the content is expanded, it affects every level of nesting inside that argument.
- Compact mode (Section 11) uses `\vadjust{\kern}` to insert space below the current line and an invisible `\vrule` to reserve height above the line. Both mechanisms operate outside TikZ.